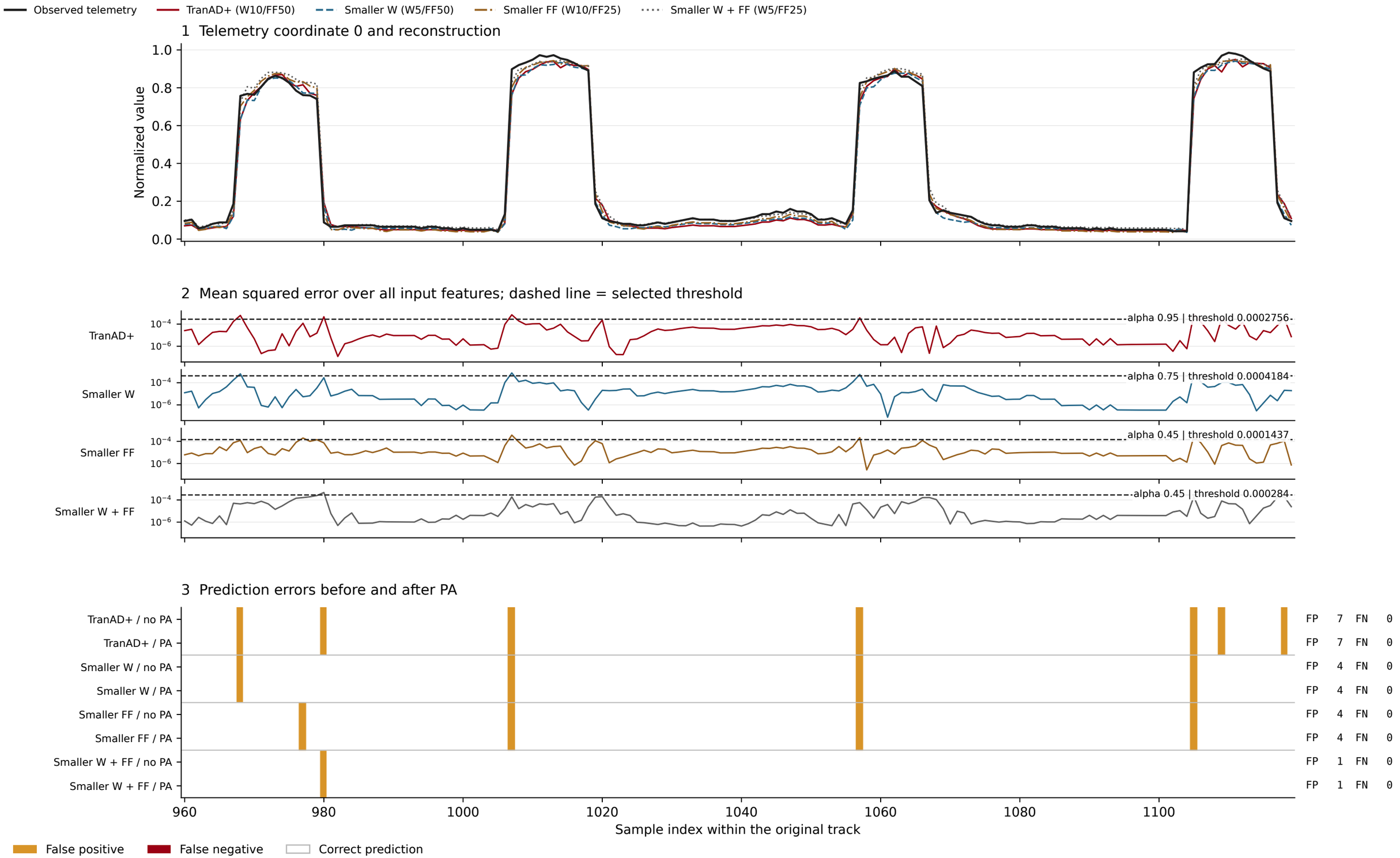
1. SMAP / E-13 / BF200

Nominal interval: false-alarm comparison

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 3/3 seeds (BF)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

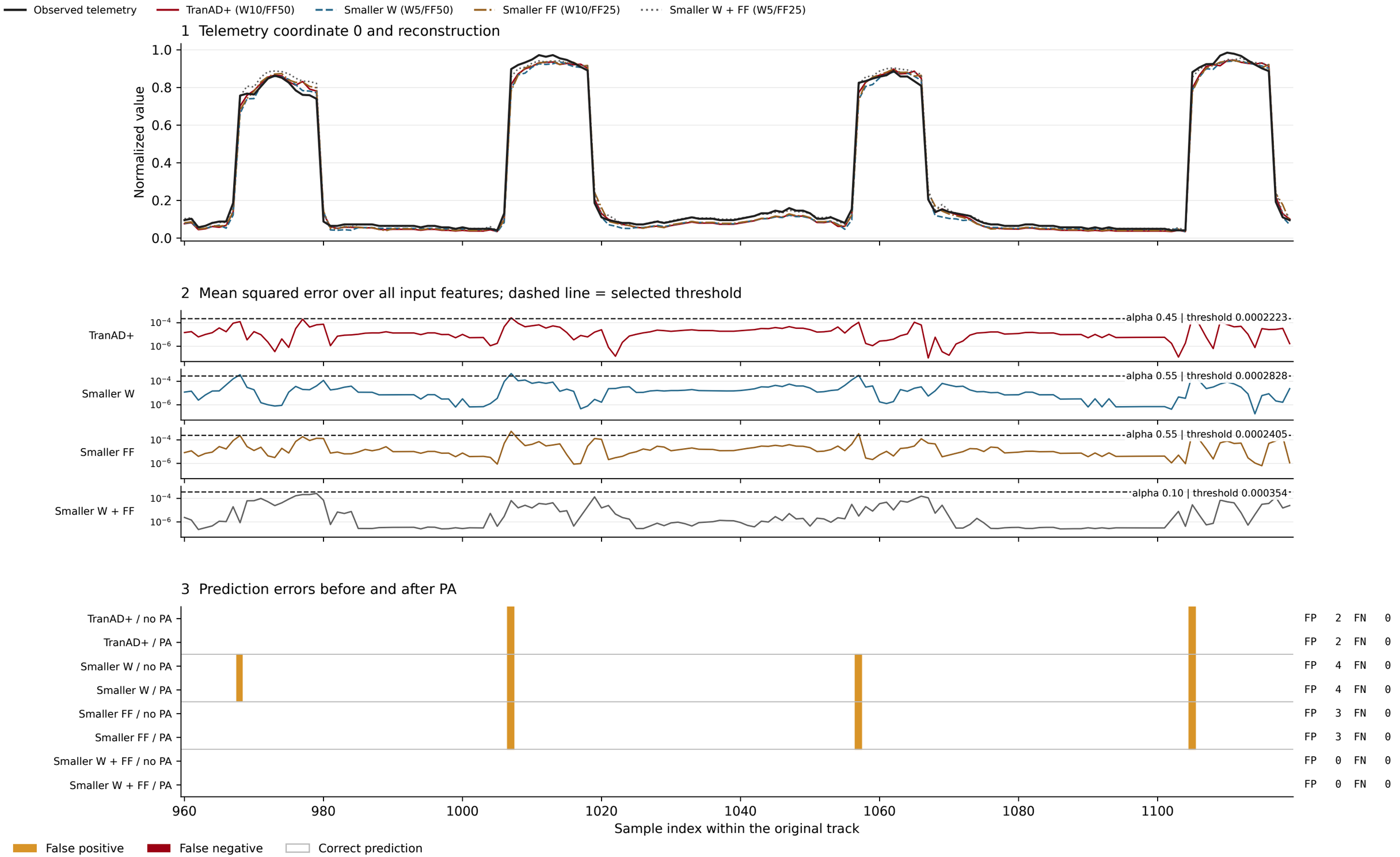
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 96.2%, 98.7%, 98.0%, 97.3%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

1. SMAP / E-13 / POT

Nominal interval: false-alarm comparison

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 3/3 seeds (BF)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

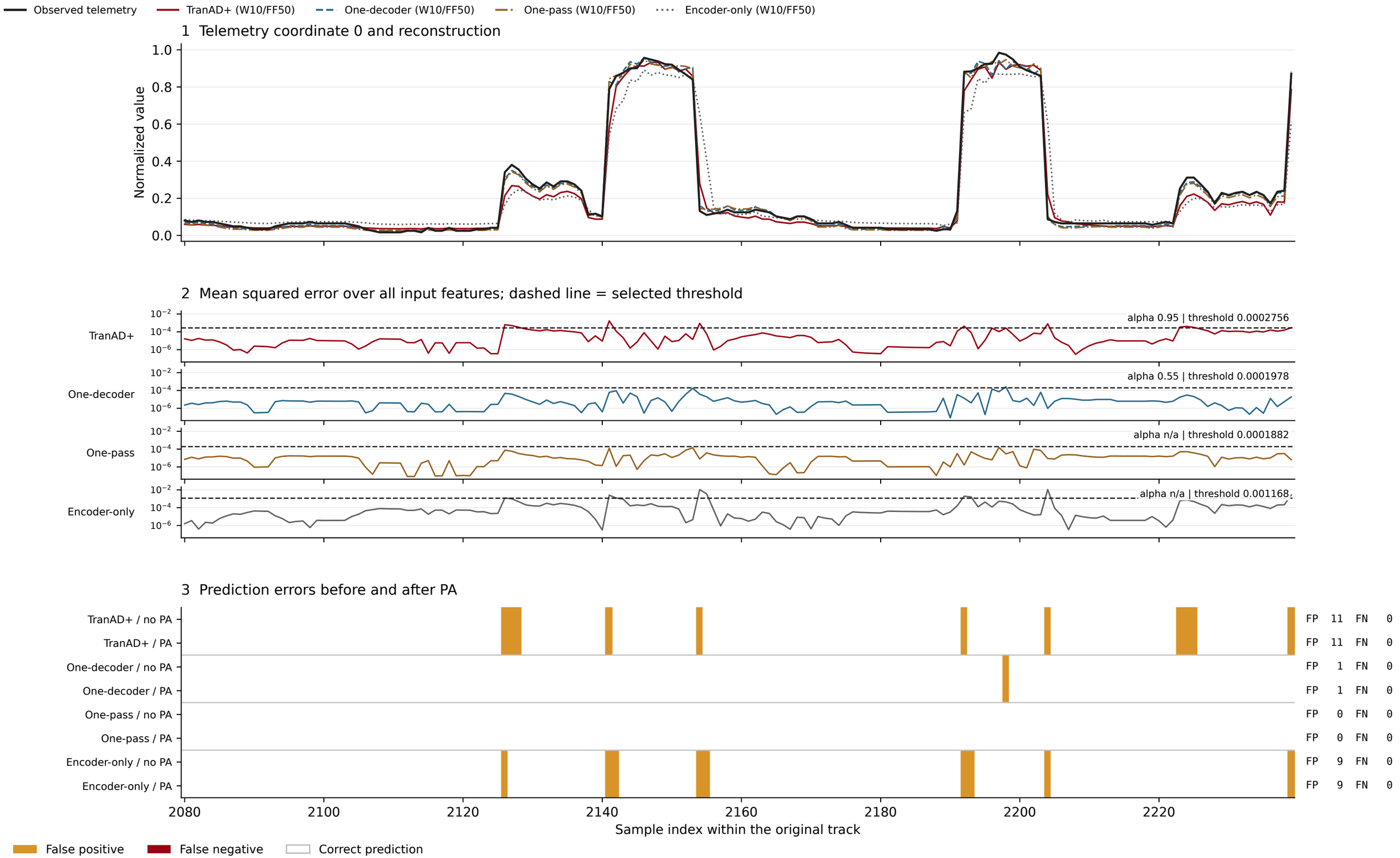
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 97.9%, 98.9%, 97.9%, 98.6%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

2. SMAP / E-13 / BF200

Nominal interval: false-alarm comparison

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (BF)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

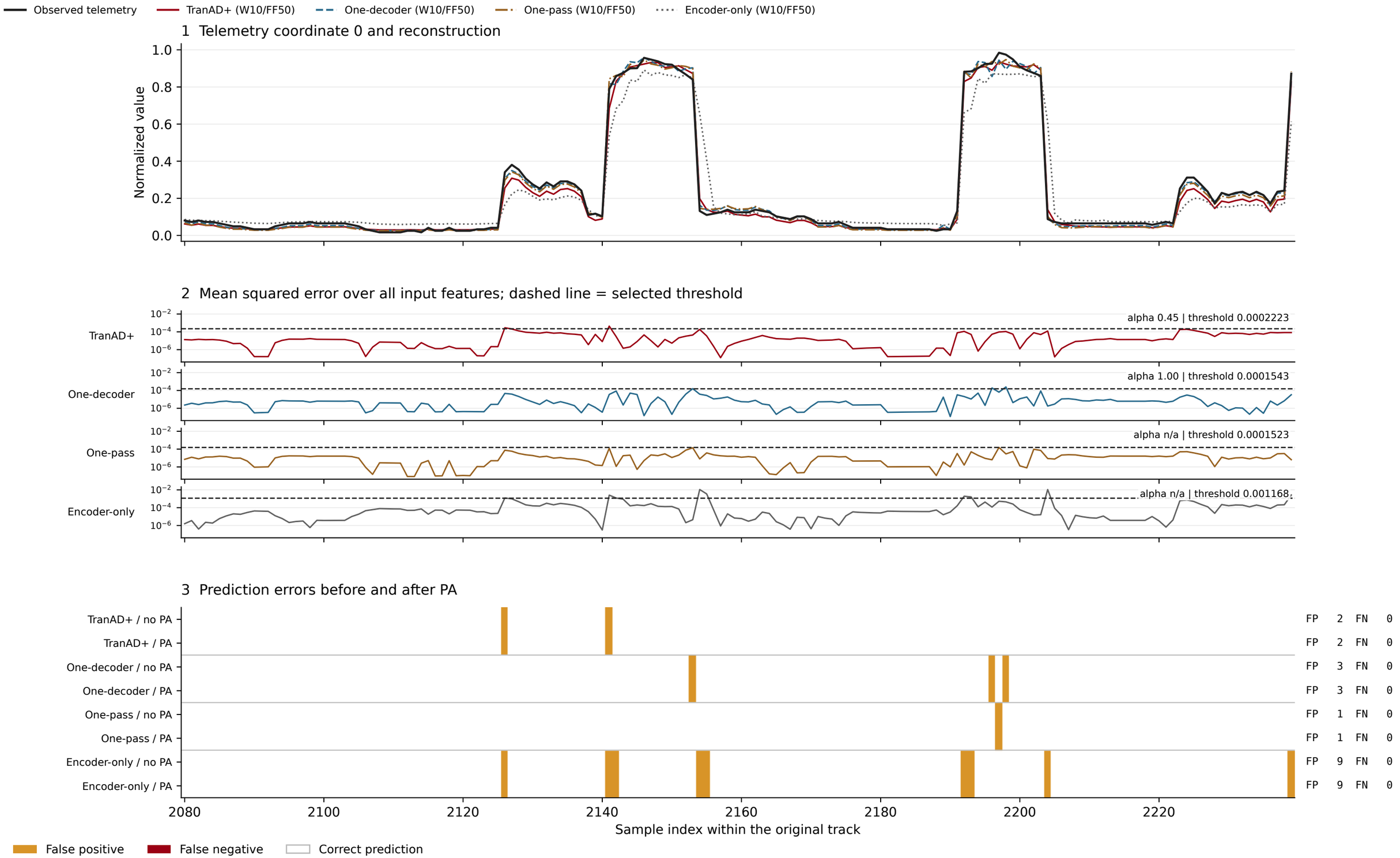
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 97.2%, 98.0%, 98.9%, 99.8%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

2. SMAP / E-13 / POT

Nominal interval: false-alarm comparison

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (BF)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

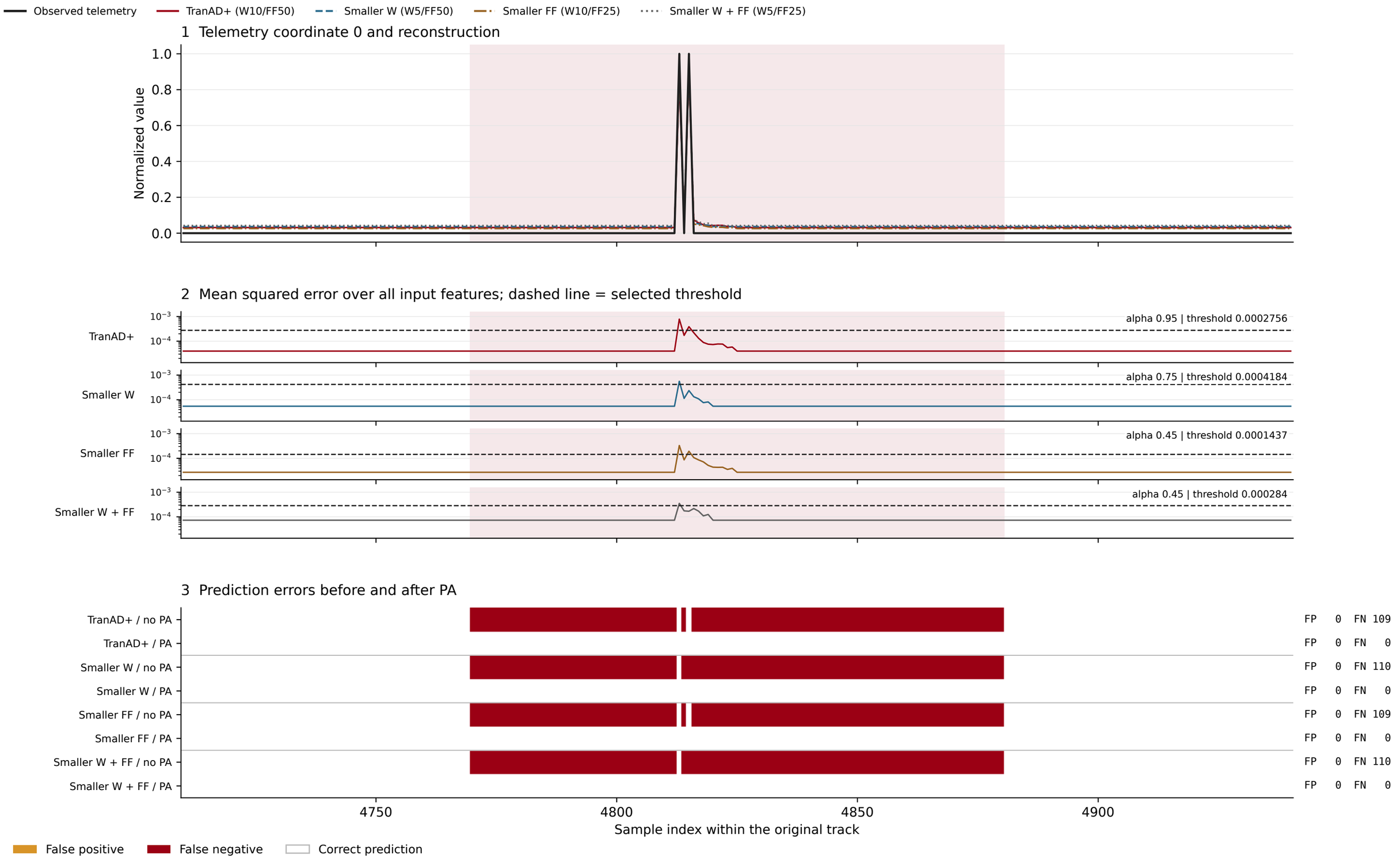
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 98.1%, 96.9%, 98.9%, 99.8%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

3. SMAP / P-4 / BF200

Anomalous interval: fewer direct hits can retain PA credit

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (BF)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

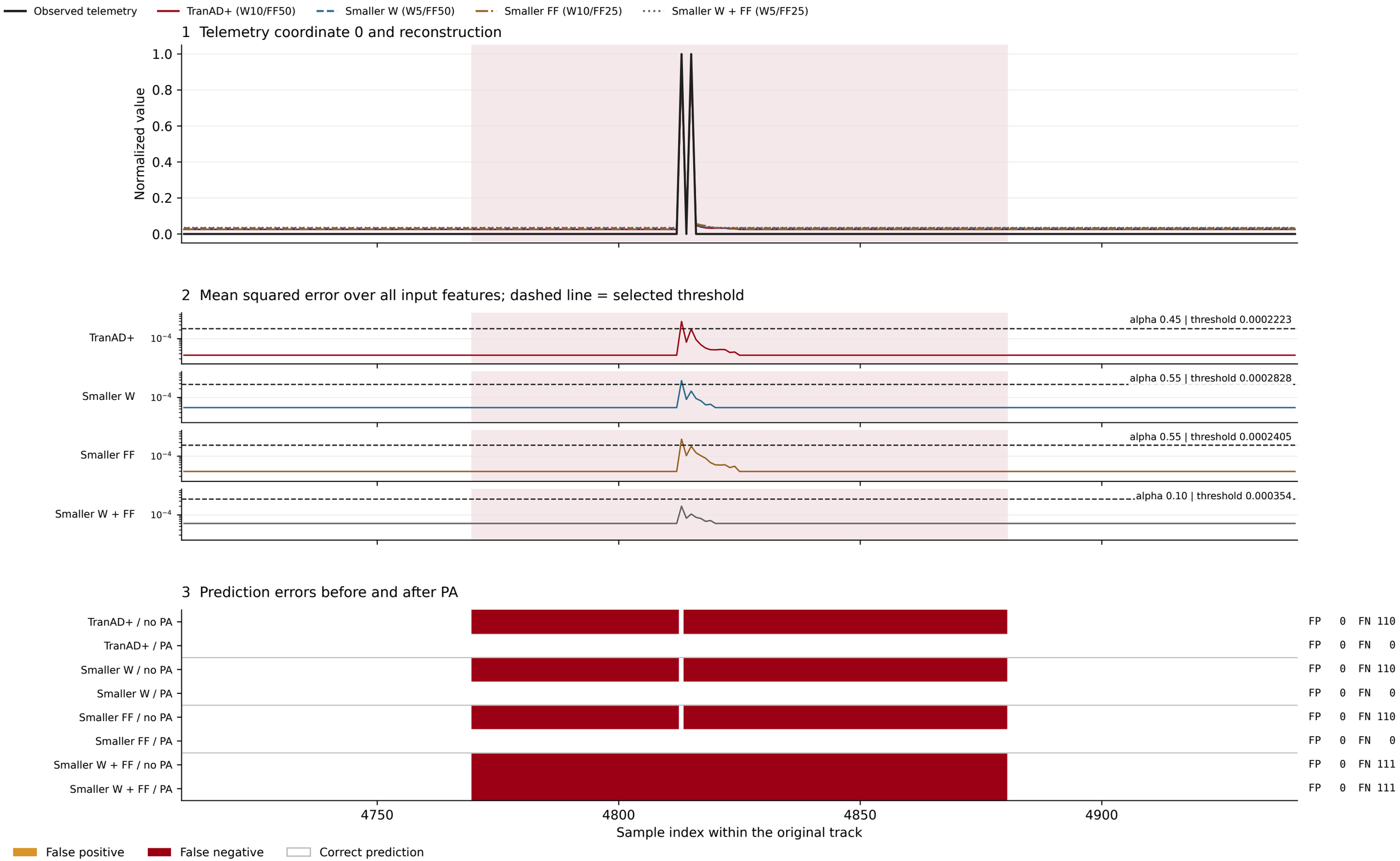
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 99.3%, 99.5%, 98.9%, 99.6%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

3. SMAP / P-4 / POT

Anomalous interval: fewer direct hits can retain PA credit

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (BF)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

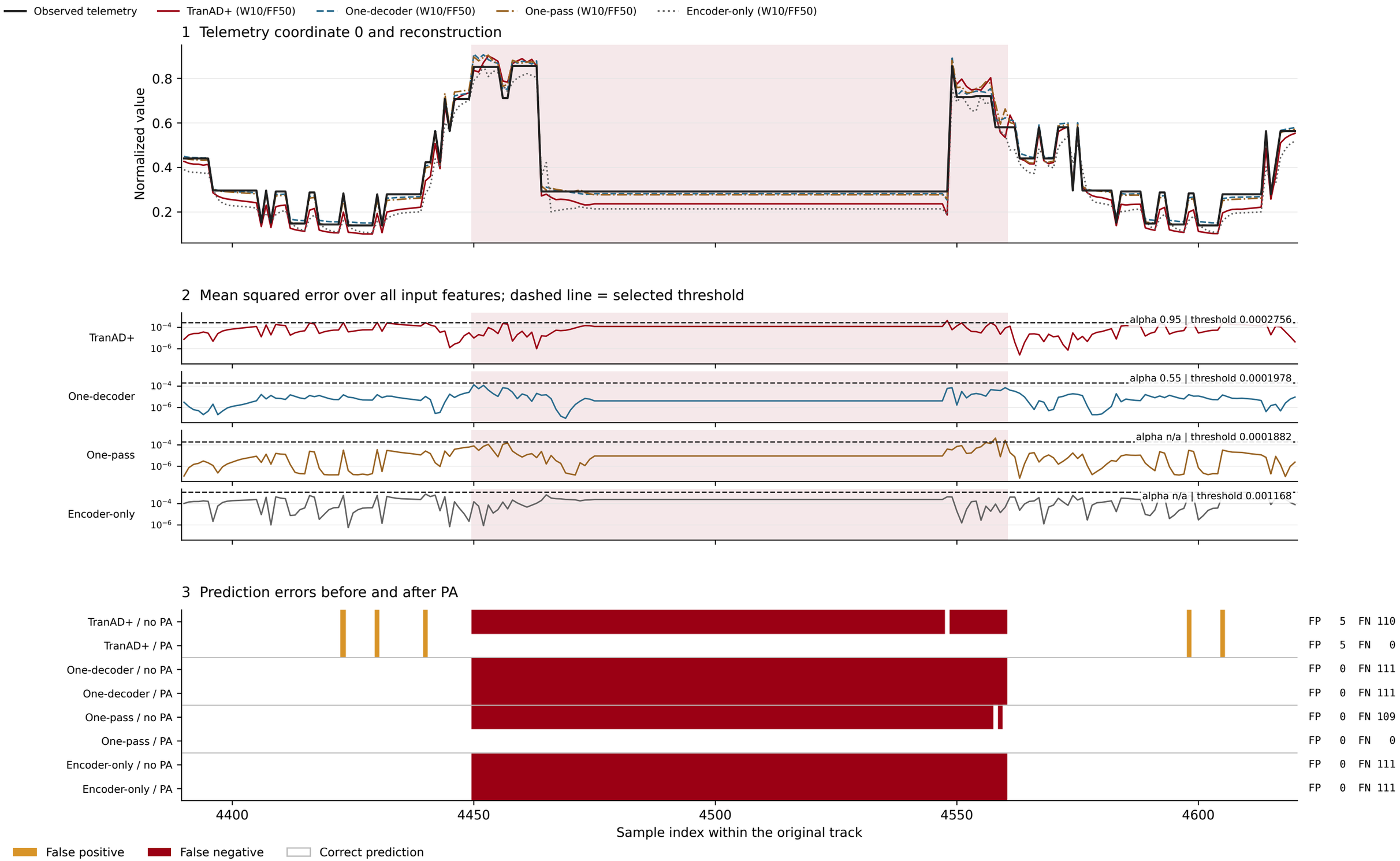
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 99.5%, 99.5%, 98.9%, 99.6%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

4. SMAP / A-2 / BF200

Counterexample: simpler architecture can miss an event

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (BF)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

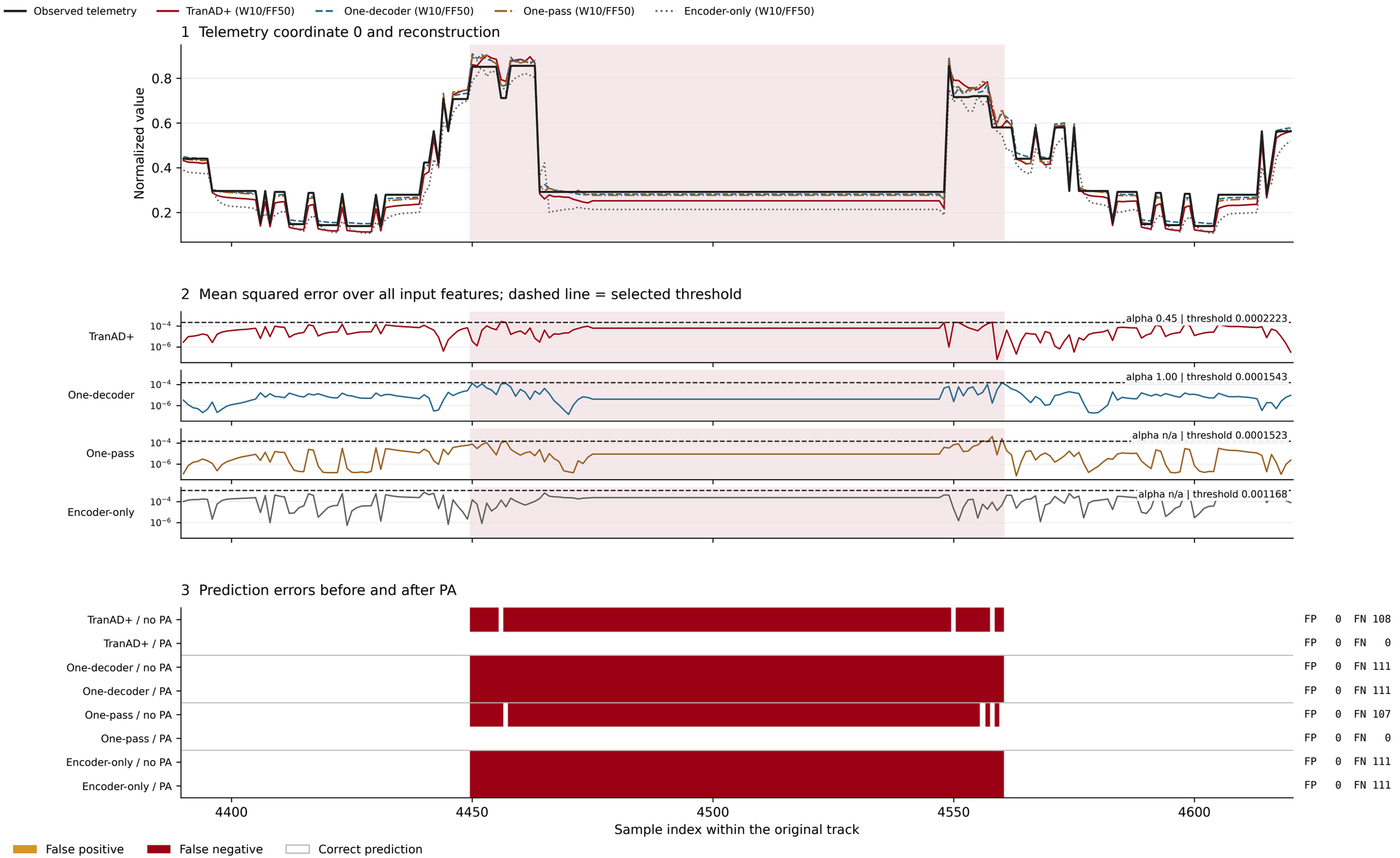
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 99.3%, 98.2%, 99.1%, 99.8%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

4. SMAP / A-2 / POT

Counterexample: simpler architecture can miss an event

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (BF)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

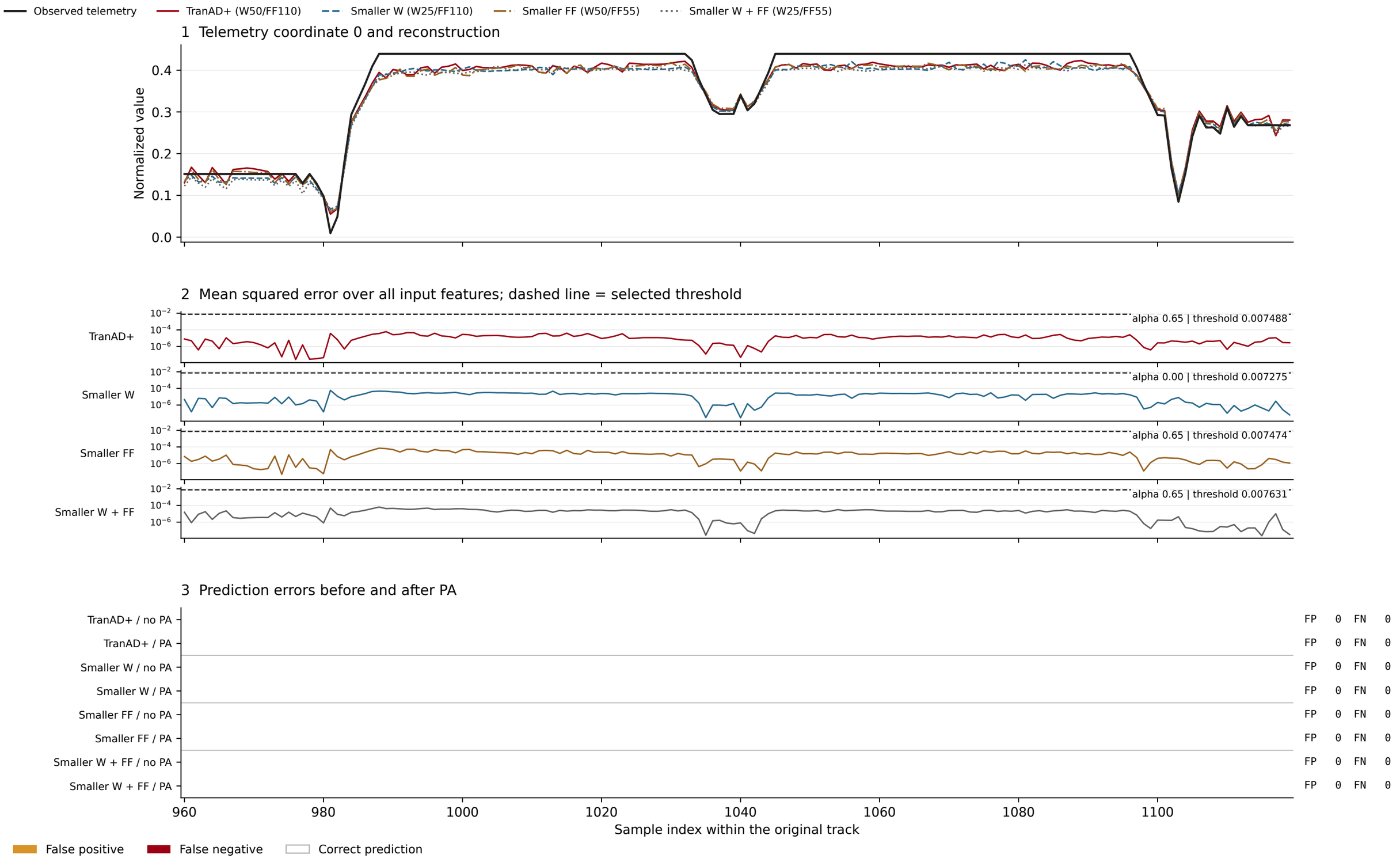
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 99.4%, 98.4%, 99.1%, 99.8%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

5. MSL / M-3 / BF200

Nominal interval: false-alarm comparison

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 3/3 seeds (POT)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

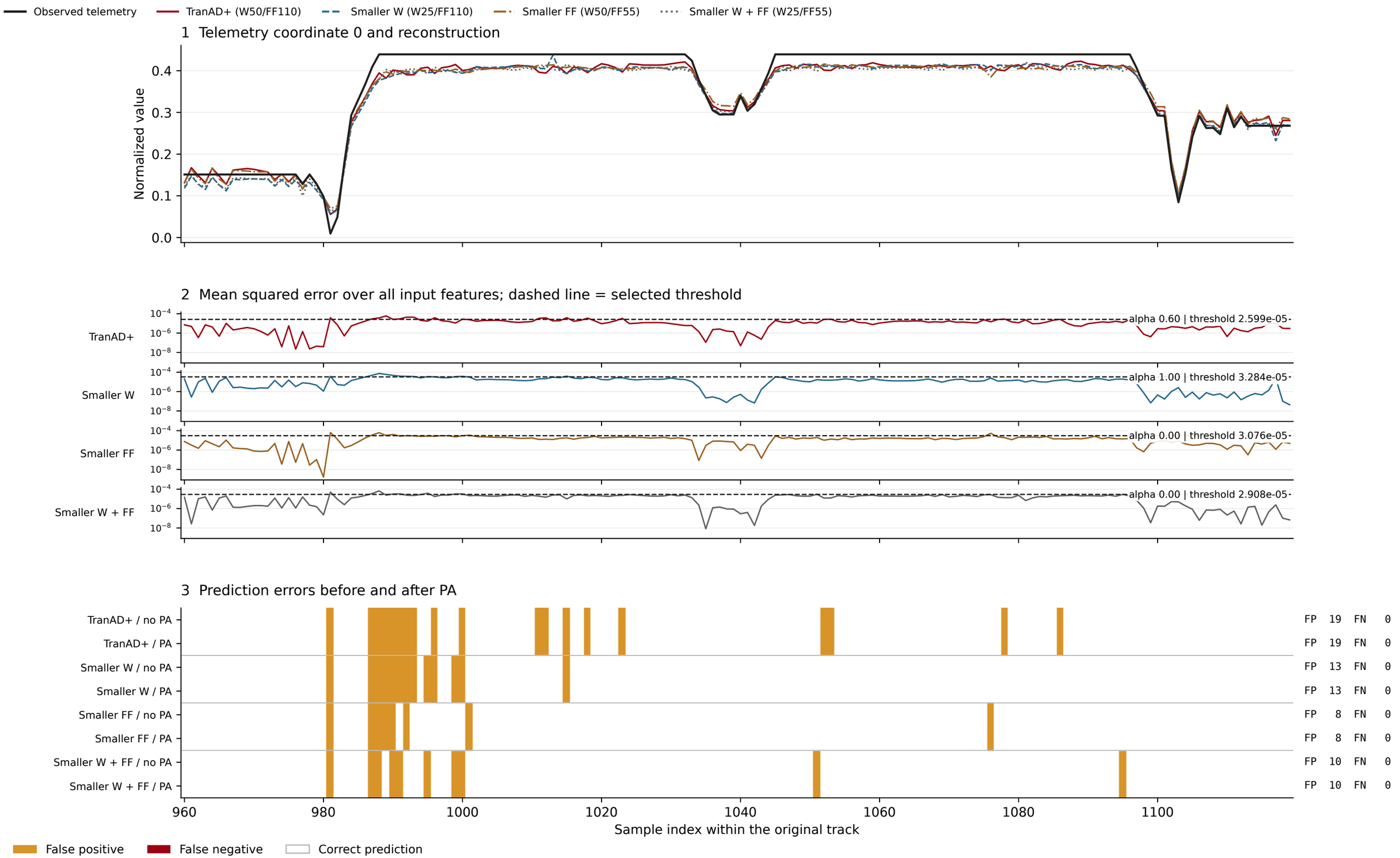
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 99.2%, 99.5%, 99.3%, 99.6%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

5. MSL / M-3 / POT

Nominal interval: false-alarm comparison

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 3/3 seeds (POT)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

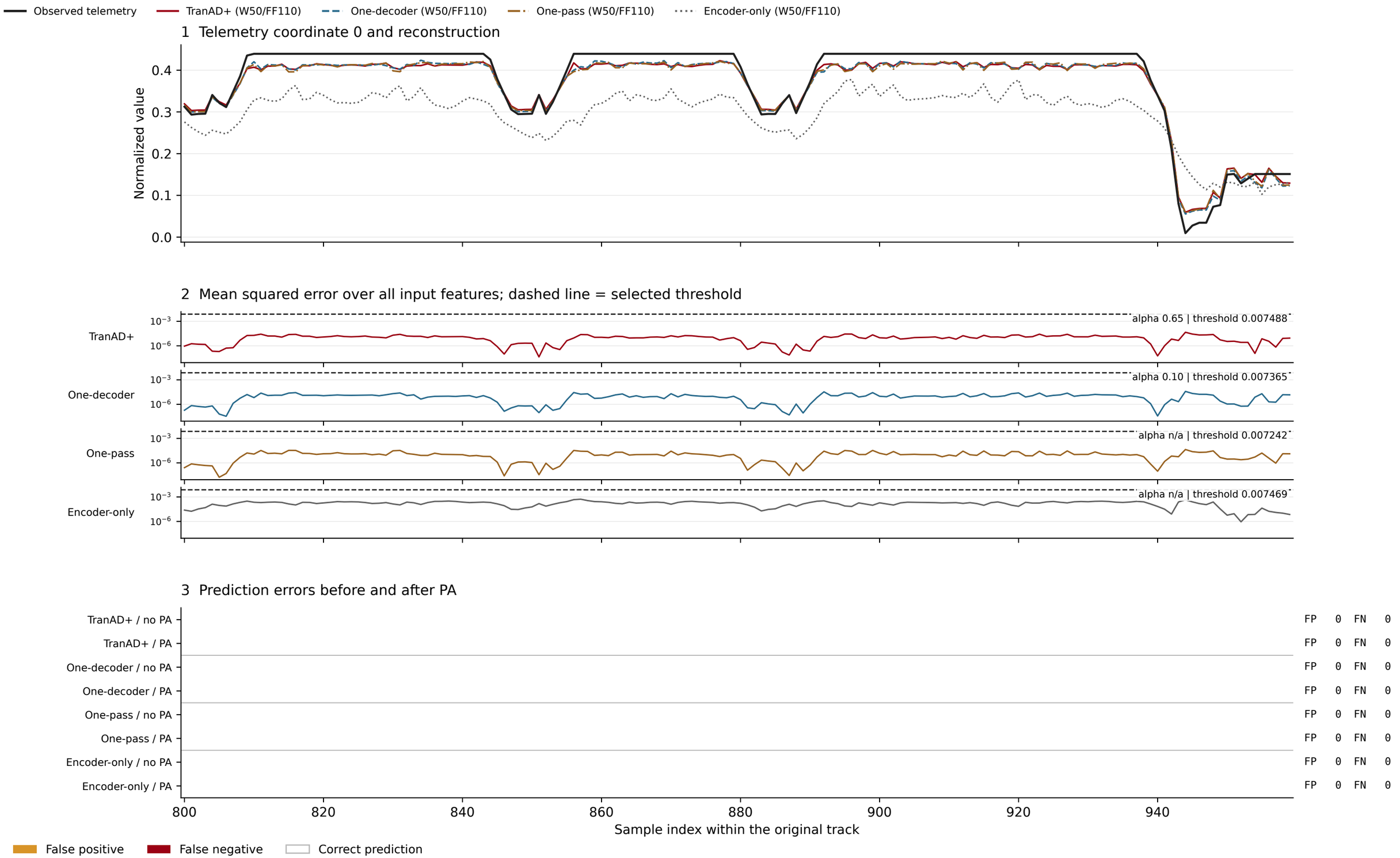
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 99.2%, 97.7%, 99.2%, 99.7%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

6. MSL / M-3 / BF200

Nominal interval: false-alarm comparison

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (POT)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

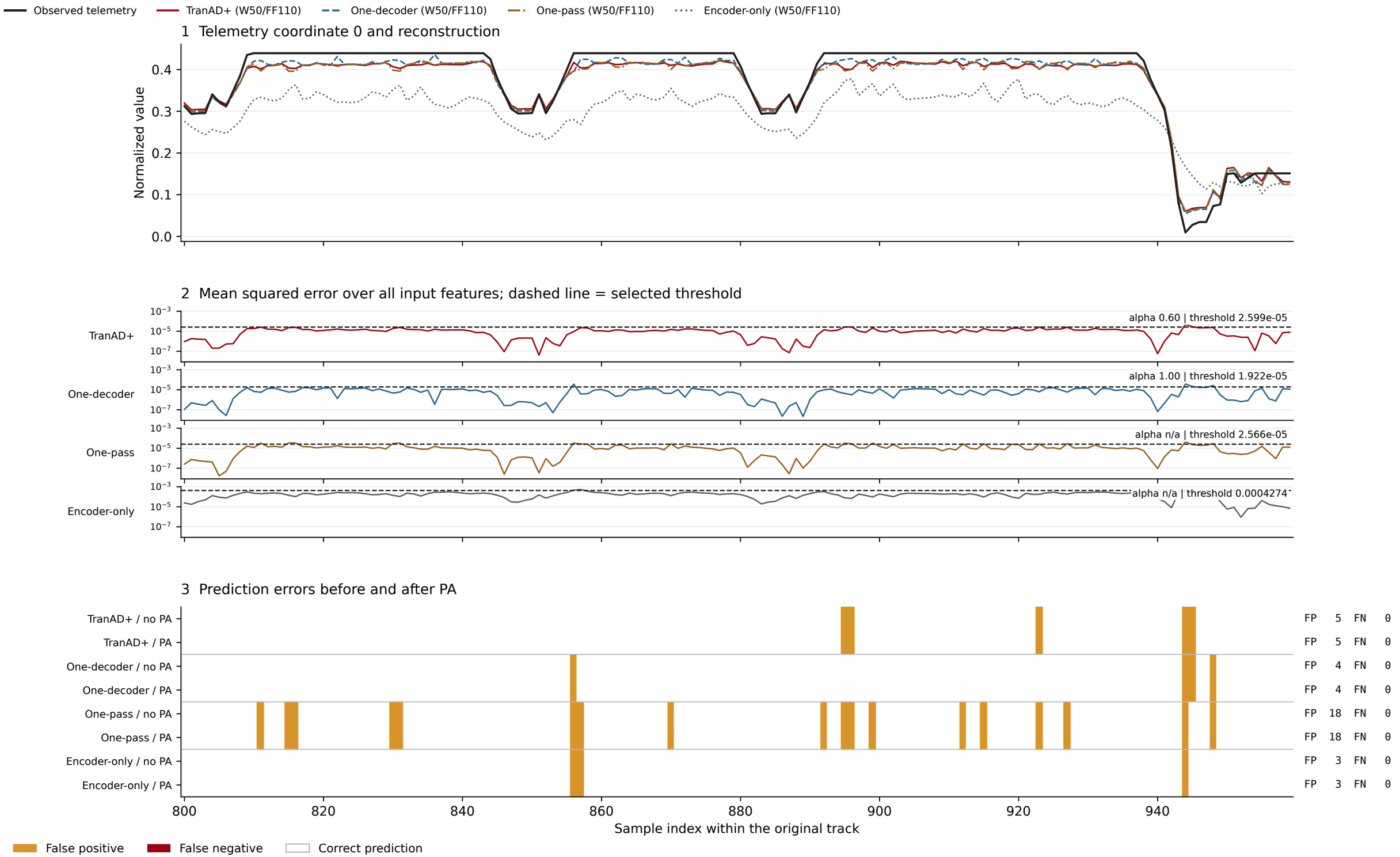
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 98.8%, 99.3%, 99.4%, 98.6%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

6. MSL / M-3 / POT

Nominal interval: false-alarm comparison

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (POT)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

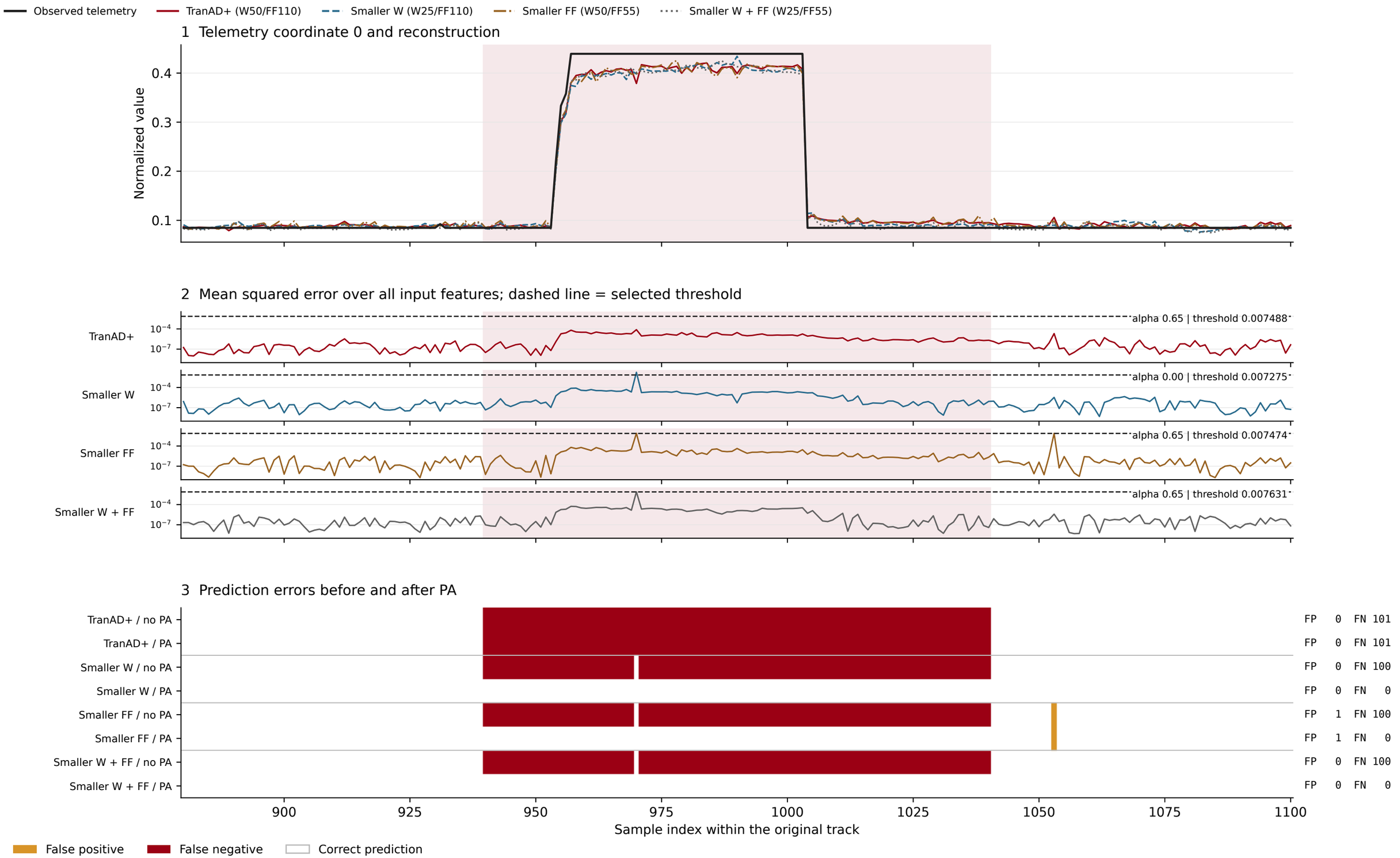
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 98.8%, 98.5%, 99.4%, 98.6%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

7. MSL / M-7 / BF200

Anomalous interval: threshold selection changes event coverage

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (POT)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

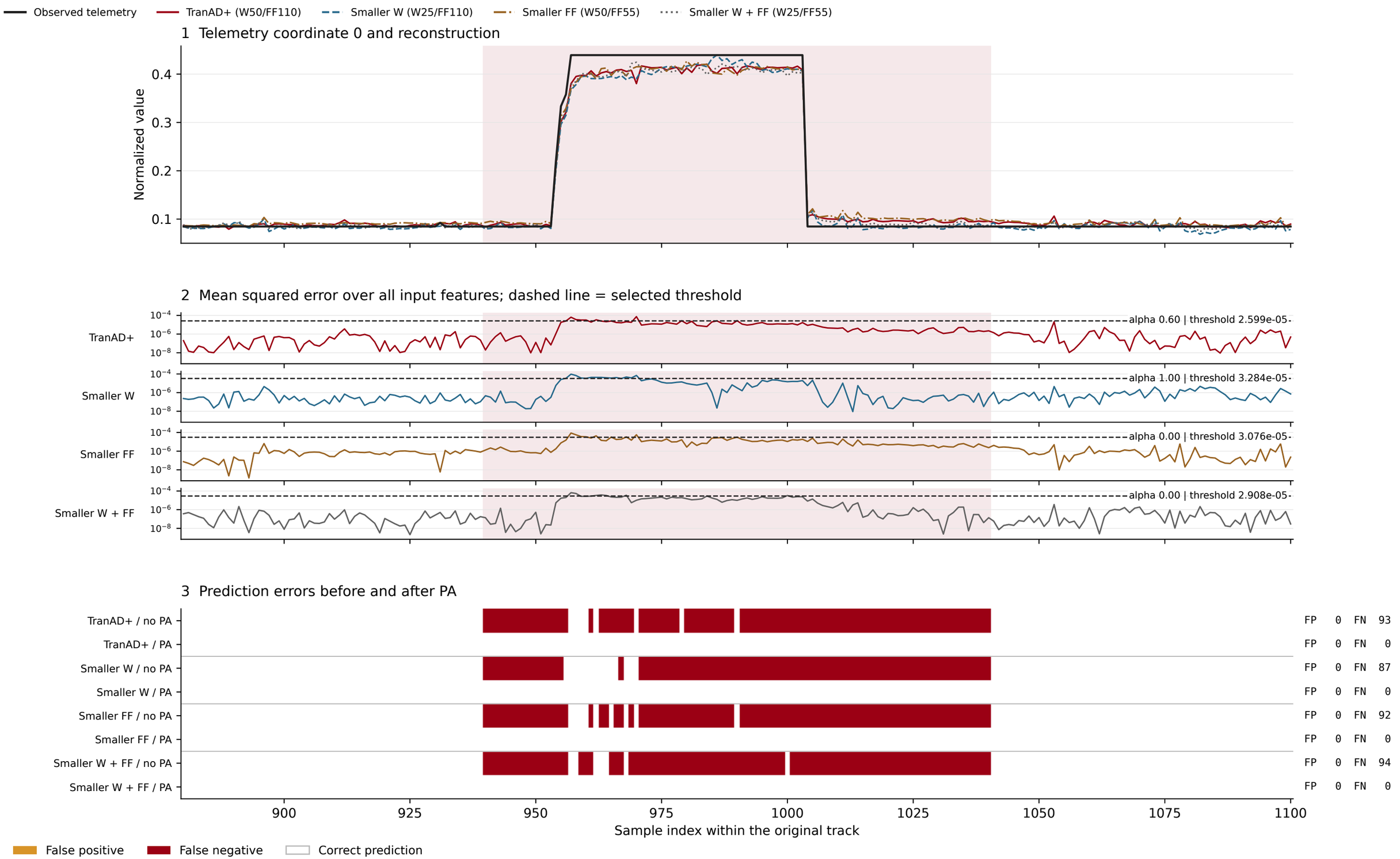
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 96.4%, 6.8%, 7.5%, 13.5%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

7. MSL / M-7 / POT

Anomalous interval: threshold selection changes event coverage

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 2/3 seeds (POT)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

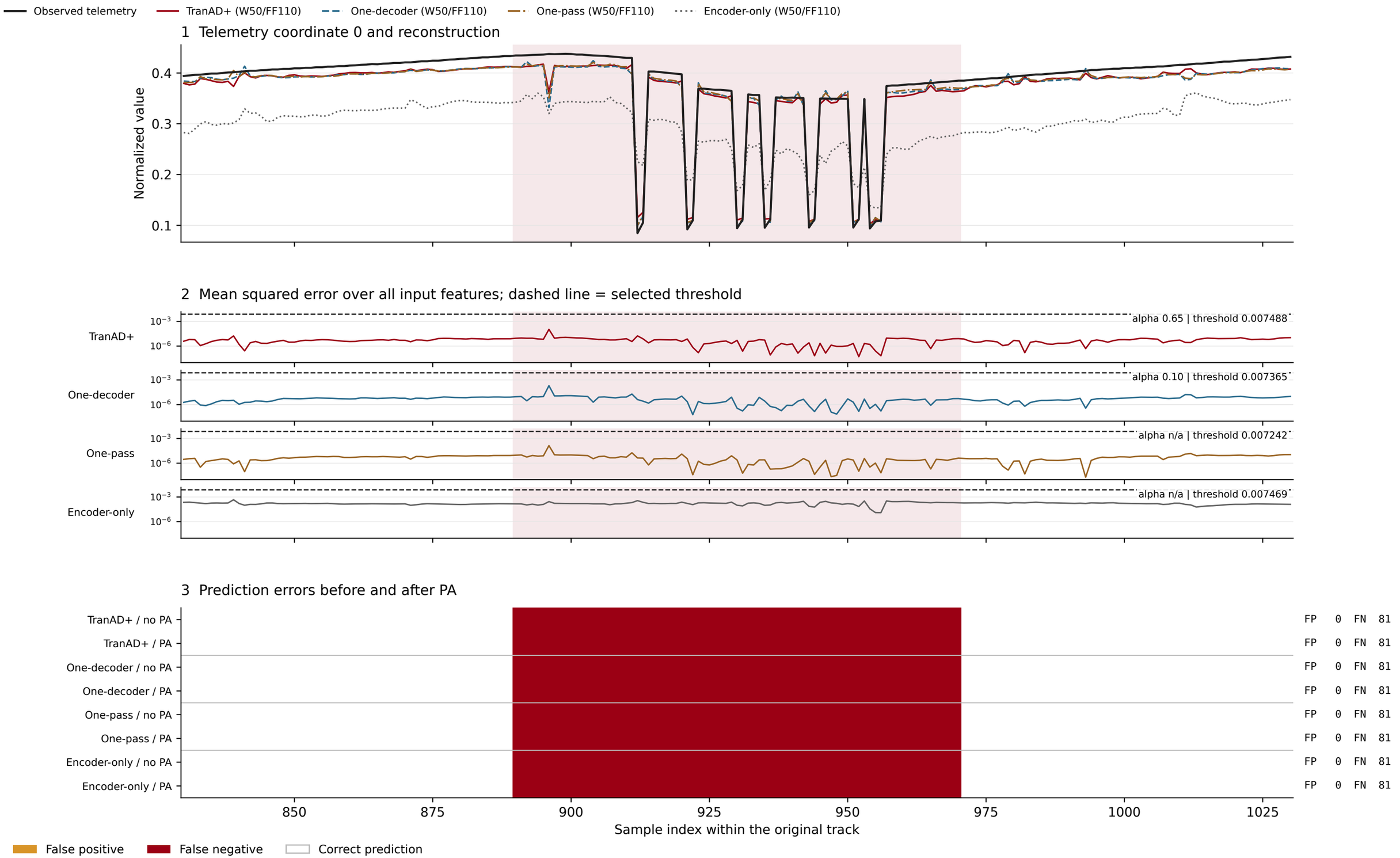
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 96.5%, 96.4%, 95.5%, 98.8%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

8. MSL / T-9 / BF200

Counterexample: simpler architecture can miss an event

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 1/3 seeds (POT)



Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

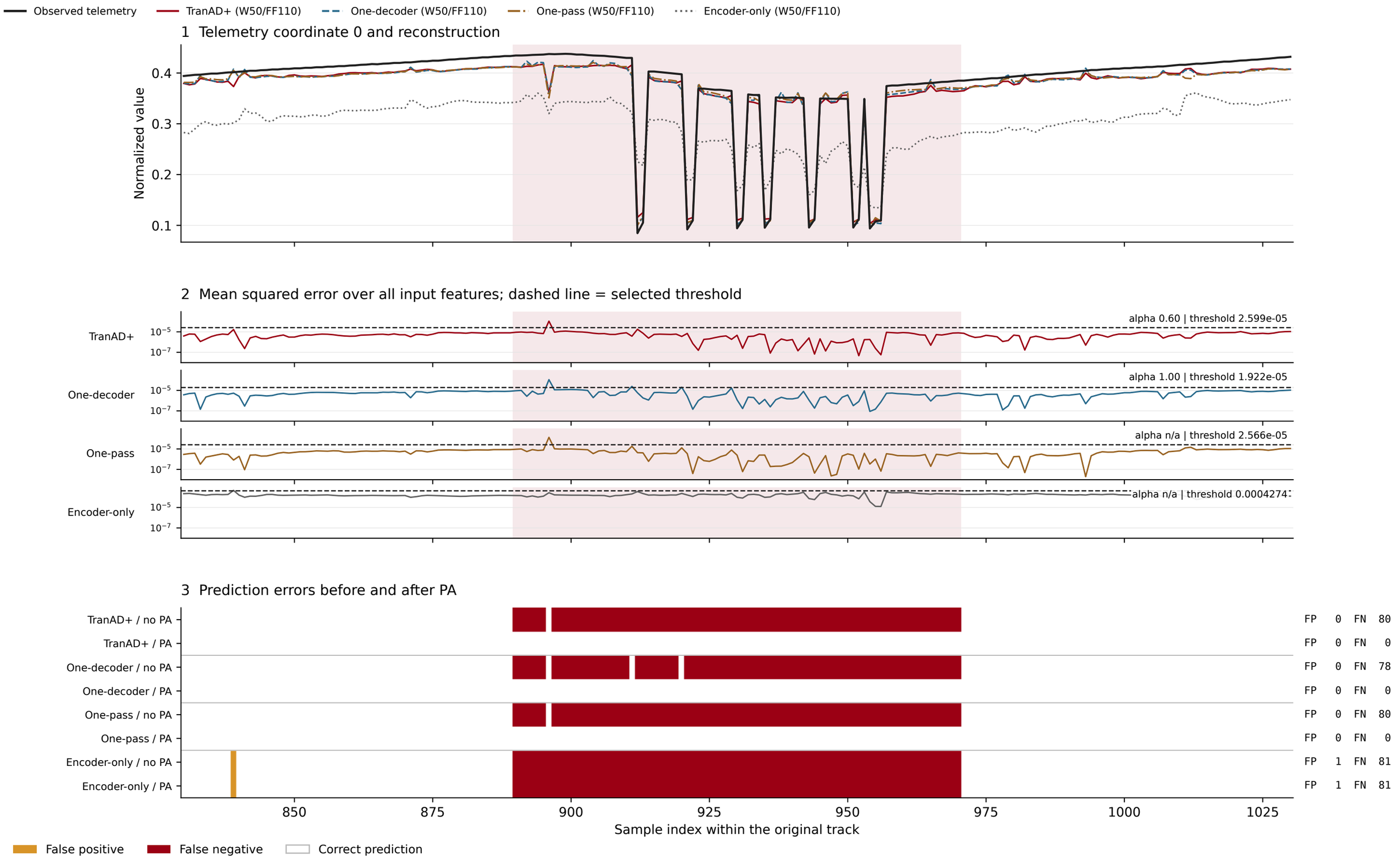
Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 97.7%, 99.0%, 99.3%, 98.5%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

8. MSL / T-9 / POT

Counterexample: simpler architecture can miss an event

Original saved alpha and threshold for each model | seed2026 | selection category agrees in 1/3 seeds (POT)



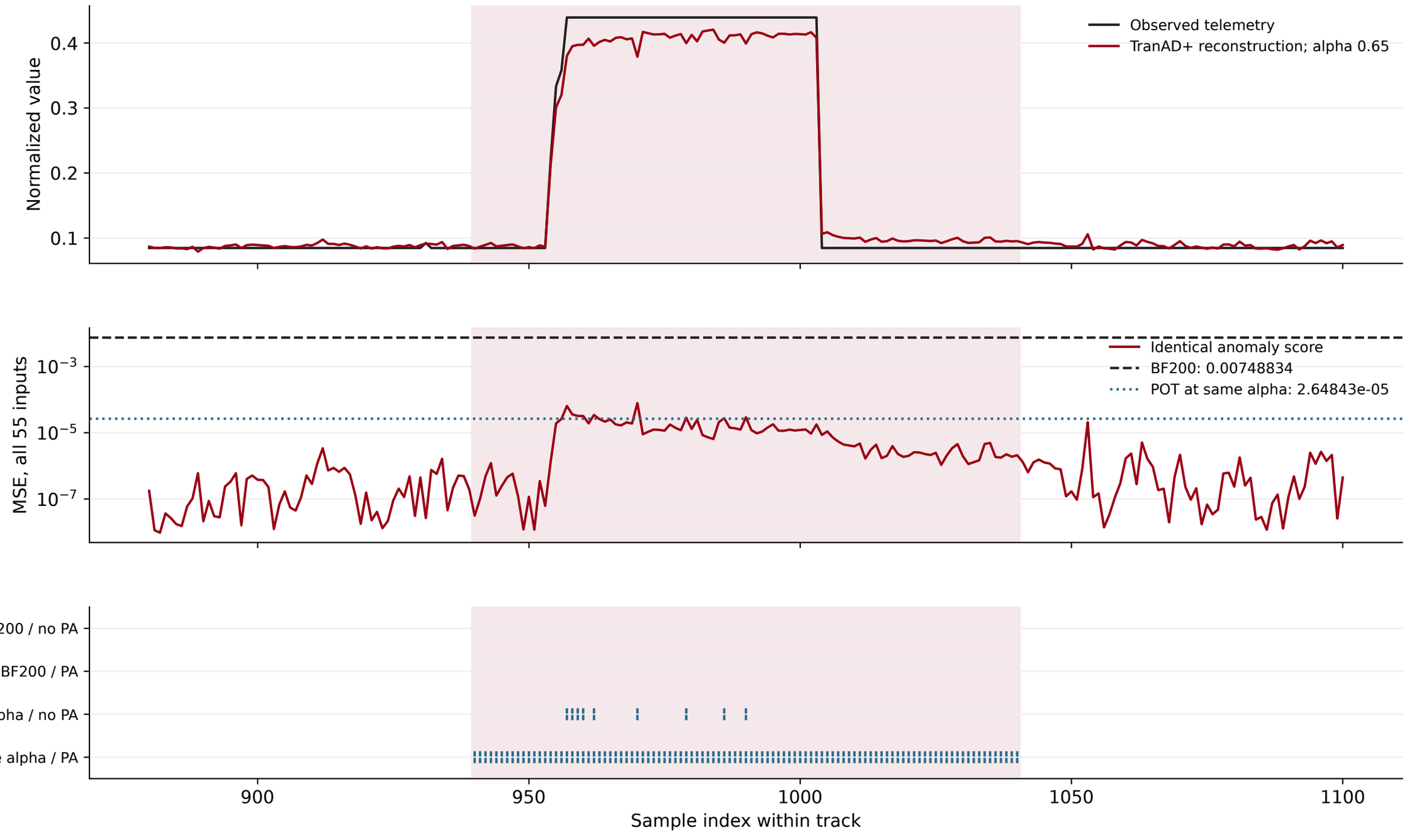
Pale shading: supplied positive labels. PA is applied on the full original stream before cropping; false positives remain unchanged.

Top shows one telemetry coordinate; score uses all inputs. Coordinate-0 share of displayed MSE: 97.8%, 97.4%, 99.3%, 98.5%.

Illustrative, outcome-selected examples; candidate lists and selection rules are saved. Anonymous channel IDs do not establish a physical fault cause.

MSL / M-7: same model, same alpha, same score

Only the threshold changes. This control does not retrain or optimize another alpha.

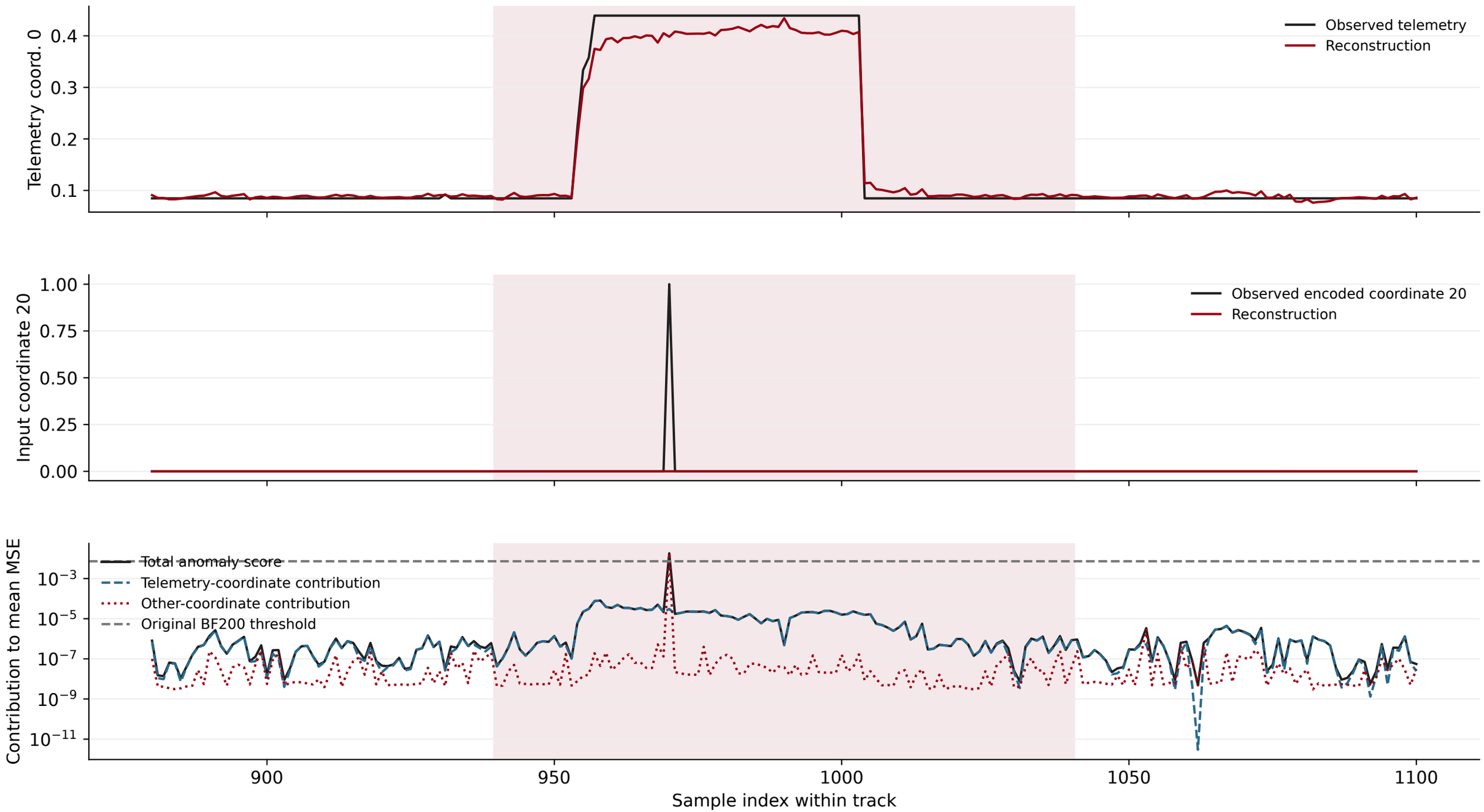


Entire MSL test stream, alpha 0.65: PA F1 40.71% -> 84.97% | raw F1 5.74% -> 15.41%

Continuous-score ROC-AUC is unchanged: 61.00%. BF200 first grid step: 0.00748833.

MSL / M-7: which input creates the BF200 alarm?

W25 / FF110; original BF200 alpha=0.0. Coordinate 0 is the telemetry; others encode command-related inputs.



At sample 970, telemetry contributes 0.17% of the score; other coordinates contribute 99.83%.

This alarm lies inside the annotated event. Its feature contribution does not identify a physical command fault or prove that it is spurious.